Universitatea Babeş-Bolyai, Facultatea de Matematică şi Informatică

Secţia: Informatică engleză, Curs: Dynamical Systems

Laboratory 3. Linear Nonhomogeneous Differential Equations

Exercise 1.

a) Find the general solution to $x' + x = \frac{2}{\sqrt{\pi}} e^{-t^2 - t}$.

b) Compute the primitive of e^{t^2} and then of $\frac{2}{\sqrt{\pi}} e^{-t^2}$.

c) Write the definition of the function **erf**: $\mathbb{R} \to \mathbb{R}$.

Exercise 2.

Decide whether the following statements are true or false.

a) Any solution of $x'' + 3x' + x = 1$ verifies $\lim_{t \to \infty} x(t) = 1$.

b) The solution of the IVP $x'' + 4x = 1, \quad x(0) = 5/4, \quad x'(0) = 0$ verifies $x(\pi) = 5/4$.

c) The equation $x' = 3x + t^3$ admits a polynomial solution.

Exercise 3.

Represent the graph of the following function f in an interval that contains the point where the function changes its expression (for example the interval $(-1, 5)$). In Maple you can use the option discont=true.

Decide, looking at the graph, whether f is continuous in any point or f is differentiable in any point. In your notebooks draw the graph and write the explanations.

a) $f(t) = \begin{cases} t, & t \leq 2 \\ 3 - t, & t > 2 \end{cases}$; b) $f(t) = \begin{cases} t, & 0 \leq t \leq \pi \\ \pi e^{\pi - t}, & t > \pi \end{cases}$.

Exercise 4.

a) Find the solution $\varphi(t)$ of the IVP $\quad x'' + x = t, \quad x(0) = 0, \quad x'(0) = 1$.

Plot the graph of φ on the interval $(0, \pi)$.

b) Find the solution of the IVP $\quad x'' + x = \pi e^{\pi - t}, \quad x(\pi) = \varphi(\pi), \quad x'(\pi) = \varphi'(\pi)$.

Plot its graph on the interval (π, ∞).

c) Let f be the function given in exercise 3.b).

Find the solution of the IVP $\quad x'' + x = f(t), \quad x(0) = 0, \quad x'(0) = 1$.

Plot its graph on the interval $(0, \infty)$.

d) How are the functions in a), b), and c) interconnected?

Exercise 5.

Let $\omega > 0$ and $\varphi(t, \omega)$ be the solution of the IVP

$$x'' + x = \cos(\omega t), \quad x(0) = x'(0) = 0.$$

a) Find $\varphi(t, \omega)$ and note that it is not defined in $\omega = 1$.
b) Find separately the solution when $\omega = 1$.
c) Prove that $\lim_{\omega \to 1} \varphi(t, \omega) = \varphi(t, 1)$ for any $t \in \mathbb{R}$.
d) Note that $\varphi(t, \omega)$ is bounded for each $\omega \neq 1$ fixed. Find a bound for $|\varphi(t, \omega)|$. Plot $\varphi(t, 1)$ and note that it is unbounded. Write everything in your notebooks.

Exercise 6.

Let $\alpha > 0$ and $\varphi(t, \alpha)$ be the solution of the IVP

$$x'' - 4x = e^{\alpha t}, \quad x(0) = x'(0) = 0.$$

a) Find $\varphi(t, \alpha)$ and note that it is not defined in $\alpha = 2$.
b) Find separately the solution when $\alpha = 2$.
c) Compute $\lim_{\alpha \to 2} \varphi(t, \alpha)$ for each $t \in \mathbb{R}$.

Exercise 7.

The motion of a spring-mass system is given by the IVP

$$x'' + x = f(t), \quad x(0) = 5, \quad x'(0) = 0,$$

where $f(t) = \begin{cases} t, & t \in [0, \pi/2) \\ \pi - t, & t \in [\pi/2, \pi] \\ 0, & t \in (\pi, \infty) \end{cases}$. Find the solution of this IVP and plot its graph.

Exercise 8.

Find the solution of each of the following IVPs. Plot the planar curve with the obtained parametric equations.
a) $x' = -2x, \quad y' = -3y, \quad x(0) = 1, \quad y(0) = 1.$
b) $x' = -2x, \quad y' = 3y, \quad x(0) = 1, \quad y(0) = 1.$
c) $x' = -y, \quad y' = 4x, \quad x(0) = 1, \quad y(0) = 1.$
d) $x' = -x + 3y, \quad y' = -3x - y, \quad x(0) = 1, \quad y(0) = 1.$
e) $x' = -x + 3y, \quad y' = 3x - y, \quad x(0) = 1, \quad y(0) = 1.$